

Dr. Courtney B. Watson

Astronomer | Project & Data Manager

CONTACT

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📡 NASA/ADS publication list

EDUCATION

Ph.D., Astrophysics 📅 2022-2025
📍 Boston University

M.A., Astrophysics 📅 2019-2022
📍 Boston University

B.S., Physics 📅 2015-2018
📍 Texas A&M University

SKILLS

Programming

Python ●●●●●●
HTML/CSS ●●●●●●
SQL ●●●●●●
Java ●●●●●●
LaTeX ●●●●●●

Software & Tools

Data handling/analysis (e.g. numpy, scipy, pandas...) ●●●●●●

Visualization (e.g. matplotlib, bokeh, seaborn...) ●●●●●●

Project Management (e.g. Airtable, Smartsheet, Project...) ●●●●●●

Version Control (e.g. git) ●●●●●●

AREAS OF EXPERTISE

Data Management

Project Management

Proposal Writing

Data Analysis

Data Visualization

ABOUT ME

Detail oriented, PhD-trained scientist with 7+ years of experience in data analysis and project management, strong skills in statistical analysis and visualization, and proven ability to communicate findings clearly to diverse audiences. Looking to transition to industry, where I can apply my analytical, technical, and problem-solving skills to real-world challenges in data science, analytics, or related roles.

WORK HISTORY

Postdoctoral Researcher

📅 2025–now

📍 Harvard & Smithsonian CfA

Project and Data Manager for AGEL survey. This involved:

- Building Team Roadmap which summarizes active researchers, team resources, onboarding and offboarding team members, what data products were available, links to tutorials for data handling.
- Building Team Data Management Plan and Data Policies that defined best practices for data handling and organization and summarized how to access and upload to the permanent data archive which I established for the team.
- Preparation of figures for outreach, shared analytics pipelines, and reproducibility practices
- Collaborated with principal investigator/mentor to develop research plans and set project milestones
- Meeting coordination, milestone planning, and progress reporting for the international team of researchers
- Additional duties included mentoring Master's and undergraduate students, providing guidance and support in their research projects

Graduate Research Assistant

📅 2020–2025

📍 Boston University

Graduate thesis work employing space-based observations to study the thermodynamic properties of galaxy clusters and investigate the interplay between active galactic nuclei and their surrounding cluster environments. This involved:

- Developed research methodologies to ensure accurate data collection and analysis
- Designed and executed software to reduce data and analyze results for scientific studies
- Managed research projects from inception to completion
- Participated in regular lab meetings/discussions to share progress updates and receive feedback from colleagues
- Published in peer reviewed journals, gave presentations at professional conferences and outreach talks

Graduate Teaching Assistant

📅 2020–2024

📍 Boston University

Graduate TA for four different undergraduate astronomy courses across 100 and 200 level courses. This involved:

- Hosting labs/discussion sections outside of the typical lecture where I either walked them through their lab exercise and supervised their progress, or gave a lecture to started a discussion of the relevant concepts.
- Grading of lab reports or discussion assignments, weekly homework, and exams.
- Received award for Outstanding Teaching Fellow in Spring of 2024 in recognition of my efforts in organizing the TA Knowledge Database which compiled grading rubrics and answer keys from all undergraduate courses to reduce duplication of effort and reduce time spent on course prep for other graduate TAs.

Post-Baccalaureate Researcher

📅 2018–2019

📍 Yale University

Near-infrared spectroscopy of high redshift galaxy-galaxy mergers; Asteroseismic analysis of solar-cycle related changes in the sun. This involved:

- Developed research methodologies to ensure accurate data collection and analysis
- Designed and executed software to reduce data and analyze results for scientific studies
- Managed research projects from inception to completion
- Participated in regular lab meetings/discussions to share progress updates and receive feedback from colleagues